

Punith AJ

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SUMMARY

3+ years of experience in software engineering and data science, I've developed a web application integrating AI, yielding a notable 40% productivity boost. My expertise includes advanced models for semantic segmentation, object detection, and defect analysis in electric power systems. Proficient in ReactJS, Django, and ML algorithms, I consistently deliver exceptional outcomes in these domains.

EXPERIENCE

Full-Stack Developer | Compviz Technologies Private Limited | Bengaluru | December 2022– Present

AIMS | aims-kptcl.compviz.in | December 2022 – Present

- Directed the end-to-end development of a robust web application for precise defect analysis in electrical power lines and tower components. Engineered a scalable architecture using ReactJS (frontend) and Django (backend), and implemented secure user authentication via JSON Web Tokens (JWT).
- Fine-tuned advanced pretrained models (YOLO, Detectron, YOLO OBB) for object detection in thermal and RGB imagery of electrical power lines. Employed clustering algorithms (DBSCAN, Agglomerative Hierarchical Clustering) to group images and infer the phase of power line towers using gimbal yaw and pitch data. Utilized UNet and SegFormer (Hugging Face Transformers) for semantic segmentation of conductor wires. Enhanced analysis speed by 40% over manual workflows, significantly improving inspection efficiency.
- Developed automated pipelines to process and combine predictions from multiple model architectures ensuring comprehensive defect detection for each tower component.
- Designed and implemented a React Native mobile application enabling field technicians to track and resolve electrical power line defects more efficiently, leading to improved operational accuracy and digitizing technician workflows.

Feeder Analytics | kfeans.kptcl.net | February 2024 – Present

- Developed a Feeder classification system to determine power requirements based on rainfall and crop water needs, helping to estimate power demand for various time periods.
- Designed an interactive dashboard using Apache Superset (a BI tool) to classify electrical feeders for analysis and decision-making, utilizing PostgreSQL to create datasets, generate charts, and implement row-level security based on user and office linking for secure and role-specific data access.

EDUCATION

M.Tech in Digital communication | RV College Of Engineering, Bengaluru | CGPA – 8.05

SKILLS

Programming Language: Python, JavaScript

Backend: Django, REST Framework

Frontend: HTML, CSS, Bootstrap, jQuery, ReactJS, React Native

Database: PostgreSQL

AI: Machine Learning, Deep Learning, Computer Vision, NLP, GANs, Pytorch, TensorFlow, NumPy, Pandas, Scikit-Learn, OpenCV, LangChain